

01 / PROBLEM · FAILURE MODES

Why current research agents fall short.

Three obstacles to sustained autonomous research.

01 CLOSED-WORLD WORKFLOW

Observe Hypothesize Code Evaluate

× ERROR

dead end

Predefined tools and rigid paths cannot handle the unexpected.

Unexpected problems require developer intervention.

02 SINGLE TRAJECTORY

same basin

A long-lived context carries yesterday's momentum into today's search.

More iterations ≠ more exploration.

03 TREE SEARCH

UNDERSTAND → EDIT → READ CONTEXT

→

repeat

Reconstructing file context costs far more tokens than the edit itself.

Context cost forces repeated reading.

DESIGN RESPONSE

Research is open-world, non-linear, and exploratory.

ADAPT DIVERSIFY LEARN

02 / MOTIVATION

Research world

PERSISTENT / ON DISK

EVIDENCE

Knowledge + experiment ledger

Metrics, failures and original reports

CURATED

Research brief + fallible priors

Current understanding; open to revision

CODE

Retained implementation + State

Recoverable version with evidence and provenance

↓ CONTEXT RESET

Inherit experience, not prior reasoning.

03 / SYSTEM

One world. Two scientific roles.

ENVIRONMENT

Task contract

Curated data + evaluation

Starter baseline

Budget + permissions

→

RESEARCH WORLD

Verified evidence

Knowledge ledger

Research brief

Implementation + State

CONTEXT RESET

Inherit experience, not prior reasoning

→

FRESH SCIENTIST

reads world

forms hypotheses

runs experiments

reports evidence

→

META REVIEW

audits claims

scopes conclusions

curates memory

preserves optional State

Selective persistence / trajectory handoff

Audit → scope → preserve → fresh Scientist

Runtime / deterministic mechanics

Workspace isolation · runner + evaluator · logging · artifact capture · State operations

PARALLEL BREADTH isolated branches → post-hoc review → explicit adoption

04 / EXPERIENCE

Start your journey.

task.md PERSONAL.md

What to solve How you want it solved

Objectives · constraints · evaluation Preferences · research style · priorities

Define the task → Delegate research → Review evidence

Autonomous by default. Supervisable by design.

Coding agent

EXAMPLE INPUT

> Use the research-agent skill.

Follow task.md and PERSONAL.md.

Let me review the results and evidence.

05 / EVALUATION

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition

Primary score ↑

AGY direct Gemini · 0.604580

Codex direct gpt-5.6-luna ≈0.6046*

Heterogeneous agents Gemini · ≈0.6047

SciOdyssey submission 0.605936 · verified

Gemini-only SciOdyssey Cycle 2 · 0.6052

Official FM · 0.6016

Total LLM tokens (millions)

Input + output, including cache reads

* Codex is provisional. Gemini = gemini-3.7-flash. The Cycle-2 interval is a token-accounting bound. Models and protocols differ: no causal, significance or token-efficiency claim is made.

06 / VERIFIED SUBMISSION · PUBLIC VALIDATION

Verified submission · public validation

Metric	Official FM	SciOdyssey	Absolute Δ
Primary	0.6016000	0.6059363	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features × eight-seed FM ensemble · NumPy / CPU.

4 autonomous cycles 13 named experiments 0 manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* After launch, in the retained submission record.

07 / INSIGHTS

Observed behavior, not just a score.

Recovery

After evaluation, NumPy float32 values broke serialization. The Scientist repaired the writer and retained the evidence.

Audit

Reviews found leakage and sampling mismatches. META scopes evidence; it is not a formal verifier.

Open questions. One task does not establish generality. META, State and reset effects are not isolated; summaries can still lose context and anchor future work.